

Probability and Statistics

Giuliano Casale

Department of Computing, Imperial College London

Spring Term 2023/24

离散的，比如扔骰子 $\{1,2,3,4,5,6\}$



样本空间

事件

概率

- Foundations: sample spaces, events, probability
- Discrete random variables and common distributions
- Continuous random variables and common distributions
- Sums and sequences of random variables
- Jointly distributed random variables
- Statistics and estimation
- Fundamentals of simulation (**new**)

- In person lectures.
- Small group tutorials.
- Timetable:
 - Tuesdays 10-12, Huxley 308, Lecture
 - Fridays 15-16, Huxley 311, Lecture
 - Fridays 16-17, 1 hour of small group tutorials (multiple rooms)
- We will use EdStem for offline Q&A.
- Lectures will be auto-recorded on Panopto.

- Staff will monitor and reply only while the course runs. EdStem will not be used after the end of the Spring term.
- At all times, you are expected to maintain high standards of courtesy and respect on EdStem. Anonymous threads are disabled.
- If a question is judged as accidentally sharing information about an assessed question, it will be deleted. Please mark it as a private question if you are unsure if it can be asked.

Tutorials

- Fridays 16:00-17:00, in-person only.
- One per week, like year 1 MMTs/PMTs, starting 19-Jan.
- **Groups and mapping to rooms to be announced soon on EdStem and during the 19 January lecture.**
- Assessed questions due for release as already scheduled on Scientia, credit as per regulations (20%).
- Submissions **due on Wednesdays**, first assessed submission due on **31 January**. Submissions are individual or in pairs.

Groups for Friday tutorials

Groups formed based on surname prefix:

	<i>Surnames</i>	<i>GTA</i>	<i>When</i>	<i>Where</i>
1	Ahm-Cho	Margarita Bintsi	Fri, 4-5 pm	HXLY 217
2	Chu-Hug	Manon Flageat	Fri, 4-5 pm	HXLY 225
3	Hur-Leo	Alex Goodall	Fri, 4-5 pm	HXLY 218
4	Leu-Muh	Yiming Luo	Fri, 4-5 pm	HXLY 315
5	Ngu-Sat	Zifeng Niu	Fri, 4-5 pm	CHEM C661
6	Saw-Tur	Edward Pearce-Crump	Fri, 4-5 pm	HXLY 130
7	Und-Zou	Yaqi Zhou	Fri, 4-5 pm	HXLY 221

Scientia, <https://scientia.doc.ic.ac.uk/>

- Slides, mostly based on the Maths department's course notes.
- Notes cover most, but not all, of the course extremely well.
 - Around 50 examples and solved problems in it.
 - Some changes on the joint random variables part, especially Markov chains and multivariate distributions.
 - No simulation, covered in slides and books.
 - **Please read Chapters 2 and 4 to refresh basics of calculus and set theory.**
- Coursework and tutorial exercises.
- Miscellanea (e.g., FAQ, list of probability formulas).

- Check out the FAQ document.
- Exam in the Summer term: *Answer each of two questions.*
- You will be able to bring an A4 sheet as per DoC guidelines.
- You will find in the exam room DoC calculators.
- Correctness of solutions is the driving criteria in marking.
Adding some discussion alongside the mathematical derivations can help the marking process.
- Aim to use at least 3 digits after the comma to describe probabilities (e.g., 0.123 to say 12.3%)